In [ ]:
```
'''
Exercise 1a)
Univariate, Bivariate and Multivariate Analysis
'''
```

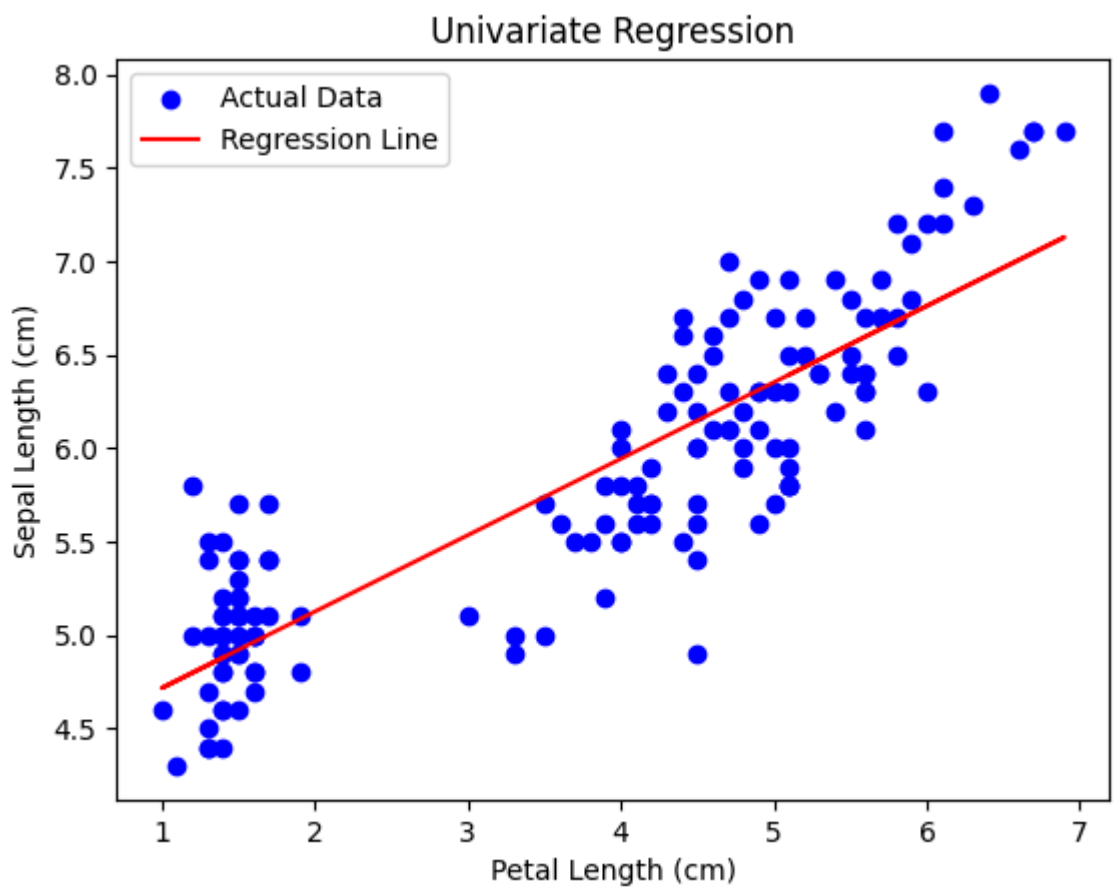
In [27]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
from mpl_toolkits.mplot3d import Axes3D
```

In [28]:
```
from sklearn.datasets import load_iris
iris = load_iris()

data = pd.DataFrame(iris.data, columns=iris.feature_names)
data['target'] = iris.target
```

In [29]:
```
y = data['sepal length (cm)']
```

In [30]:
```
# ------------------ UNIVARIATE REGRESSION ------------------
X_uni = data[['petal length (cm)']]

model_uni = LinearRegression()
model_uni.fit(X_uni, y)

y_pred_uni = model_uni.predict(X_uni)
r2_uni = r2_score(y, y_pred_uni)
```

In [31]:
```
# Plot Univariate
plt.figure()
plt.scatter(X_uni, y, color='blue', label='Actual Data')
plt.plot(X_uni, y_pred_uni, color='red', label='Regression Line')
plt.xlabel('Petal Length (cm)')
plt.ylabel('Sepal Length (cm)')
plt.title('Univariate Regression')
plt.legend()
plt.show()
```

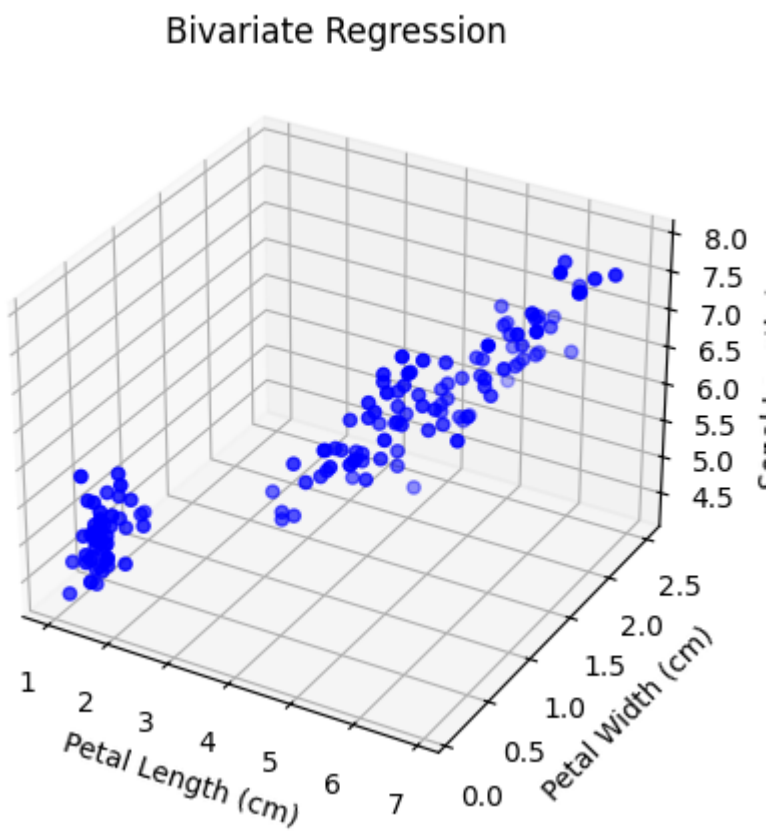


In [32]:
```
print("UNIVARIATE REGRESSION")
print("Coefficient:", model_uni.coef_)
print("Intercept:", model_uni.intercept_)
print("R-squared:", r2_uni)
```

```
UNIVARIATE REGRESSION
Coefficient: [0.40892228]
Intercept: 4.306603415047579
R-squared: 0.759954645772515
```

In [33]:
```
# ------------------ BIVARIATE REGRESSION ------------------
X_bi = data[['petal length (cm)', 'petal width (cm)']]

model_bi = LinearRegression()
model_bi.fit(X_bi, y)

y_pred_bi = model_bi.predict(X_bi)
r2_bi = r2_score(y, y_pred_bi)
```

In [34]:
```
# Plot Bivariate
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

ax.scatter(X_bi.iloc[:, 0], X_bi.iloc[:, 1], y, color='blue')

ax.set_xlabel('Petal Length (cm)')
ax.set_ylabel('Petal Width (cm)')
ax.set_zlabel('Sepal Length (cm)')
ax.set_title('Bivariate Regression')
plt.show()
```

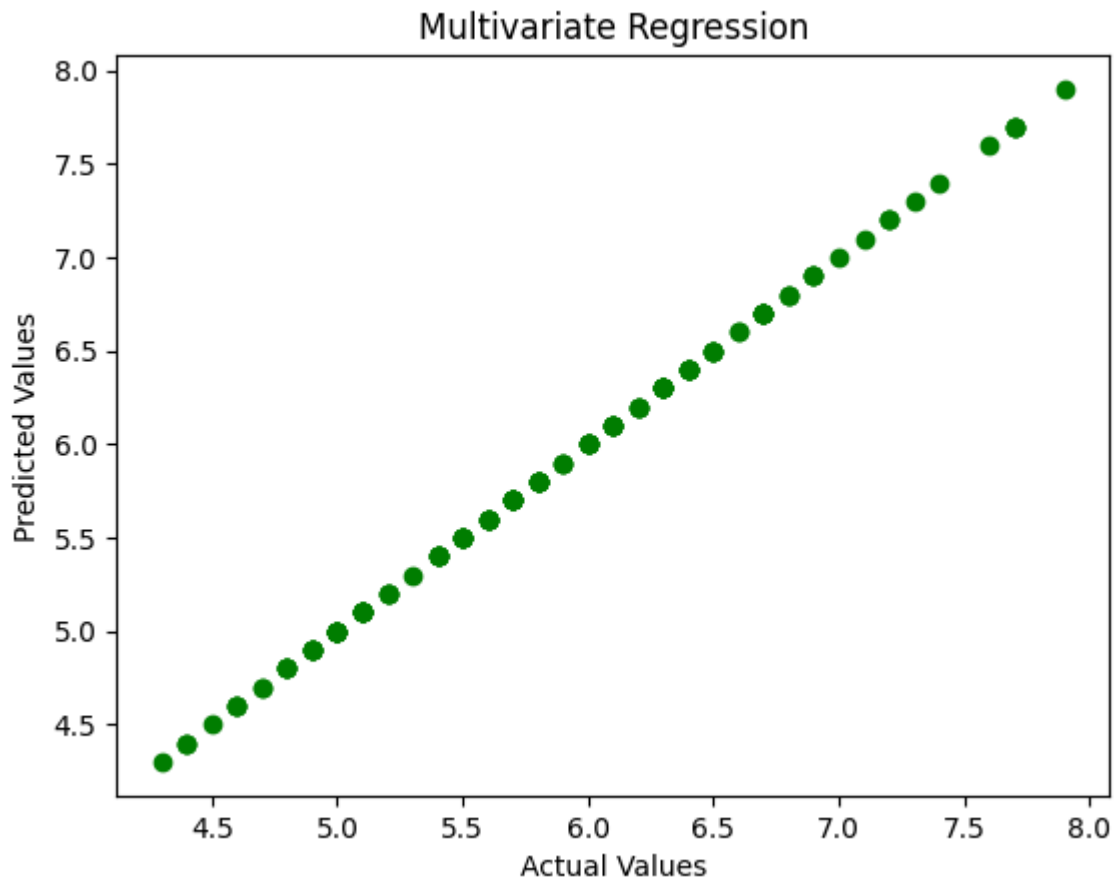


In [35]:
```
print("\nBIVARIATE REGRESSION")
print("Coefficients:", model_bi.coef_)
print("Intercept:", model_bi.intercept_)
print("R-squared:", r2_bi)
```

```
BIVARIATE REGRESSION
Coefficients: [ 0.54177715 -0.31955056]
Intercept: 4.190582428651588
R-squared: 0.7662612975425306
```

In [36]:
```
# ------------------ MULTIVARIATE REGRESSION ------------------
# Step 6: Multivariate Regression
X_multi = data.drop(columns=['target'])

model_multi = LinearRegression()
model_multi.fit(X_multi, y)

y_pred_multi = model_multi.predict(X_multi)
r2_multi = r2_score(y, y_pred_multi)
```

In [37]:
```
# Plot Multivariate
plt.figure()
plt.scatter(y, y_pred_multi, color='green')
plt.xlabel('Actual Values')
plt.ylabel('Predicted Values')
plt.title('Multivariate Regression')
plt.show()
```



In [38]:
```
print("\nMULTIVARIATE REGRESSION")
print("Coefficients:", model_multi.coef_)
print("Intercept:", model_multi.intercept_)
print("R-squared:", r2_multi)
```

```
MULTIVARIATE REGRESSION
Coefficients: [ 1.00000000e+00  6.83099711e-16  1.97657352e-17 -5.19542413e-17]
Intercept: -7.993605777301127e-15
R-squared: 1.0
```

In [ ]:
```
'''
Exercise 1b)
Data Cleaning
'''
```

In [16]:
```
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler, StandardScaler
import seaborn as sns
import matplotlib.pyplot as plt
```

In [17]:
```
df=pd.read_csv("Ex1b.csv")
```

In [18]:
```
df.head()
```

Out[18]:

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium	Heart Disease
0	70	1	4	130	322	0	2	109	0	2.4	2	3	3	Presence
1	67	0	3	115	564	0	2	160	0	1.6	2	0	7	Absence
2	57	1	2	124	261	0	0	141	0	0.3	1	0	7	Presence
3	64	1	4	128	263	0	0	105	1	0.2	2	1	7	Absence
4	74	0	2	120	269	0	2	121	1	0.2	1	1	3	Absence

In [19]:
```
df.tail()
```

Out[19]:

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium	Heart Disease
265	52	1	3	172	199	1	0	162	0	0.5	1	0	7	Absence
266	44	1	2	120	263	0	0	173	0	0.0	1	0	7	Absence
267	56	0	2	140	294	0	2	153	0	1.3	2	0	3	Absence
268	57	1	4	140	192	0	0	148	0	0.4	2	0	6	Absence
269	67	1	4	160	286	0	2	108	1	1.5	2	3	3	Presence

In [20]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 270 entries, 0 to 269
Data columns (total 14 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Age                      270 non-null    int64
 1   Sex                      270 non-null    int64
 2   Chest pain type          270 non-null    int64
 3   BP                       270 non-null    int64
 4   Cholesterol              270 non-null    int64
 5   FBS over 120             270 non-null    int64
 6   EKG results              270 non-null    int64
 7   Max HR                   270 non-null    int64
 8   Exercise angina          270 non-null    int64
 9   ST depression            270 non-null    float64
 10  Slope of ST              270 non-null    int64
 11  Number of vessels fluro  270 non-null    int64
 12  Thallium                 270 non-null    int64
 13  Heart Disease            270 non-null    object
dtypes: float64(1), int64(12), object(1)
memory usage: 29.7+ KB
```

In [21]:
```
df.isnull().sum()
```

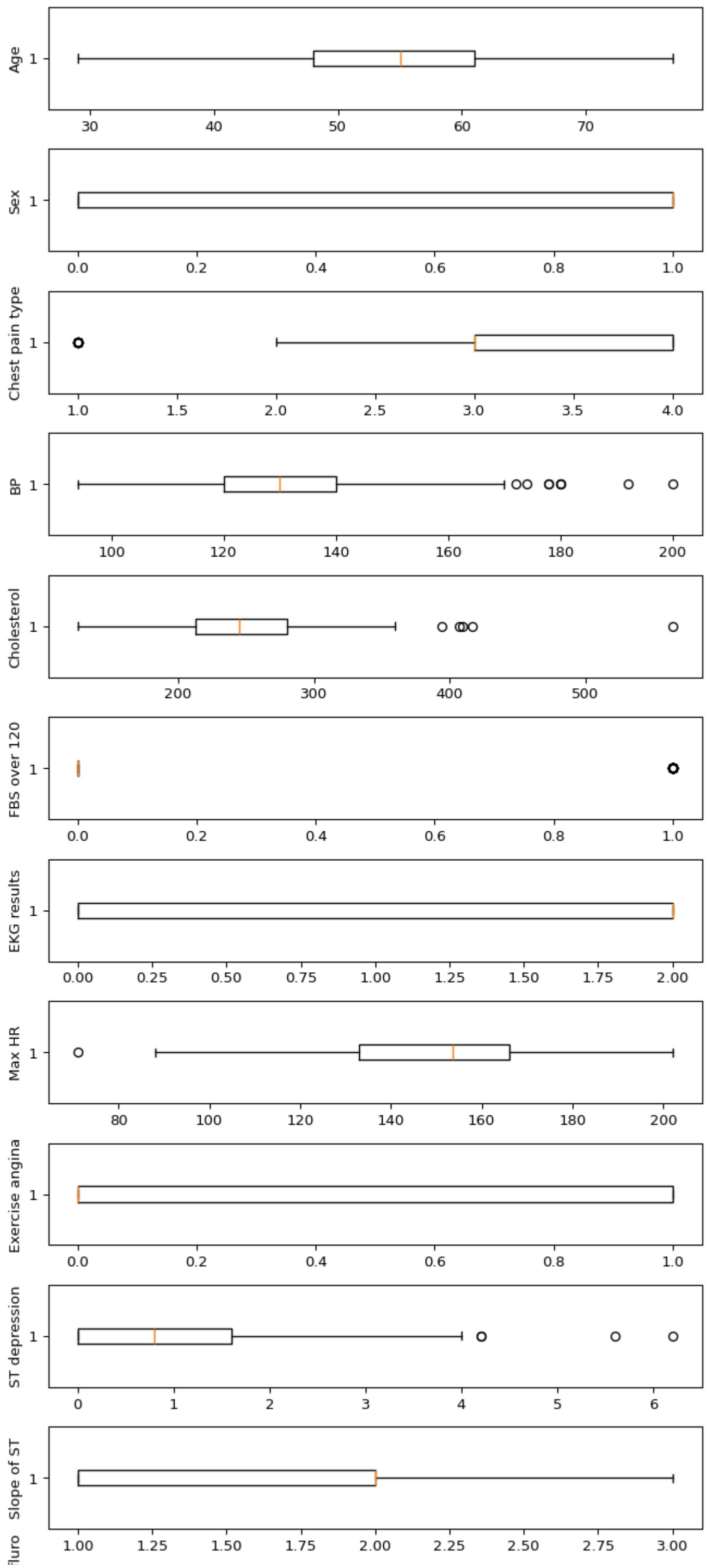
Out[21]:
```
Age                       0
Sex                       0
Chest pain type           0
BP                        0
Cholesterol               0
FBS over 120              0
EKG results               0
Max HR                    0
Exercise angina           0
ST depression             0
Slope of ST               0
Number of vessels fluro   0
Thallium                  0
Heart Disease             0
dtype: int64
```

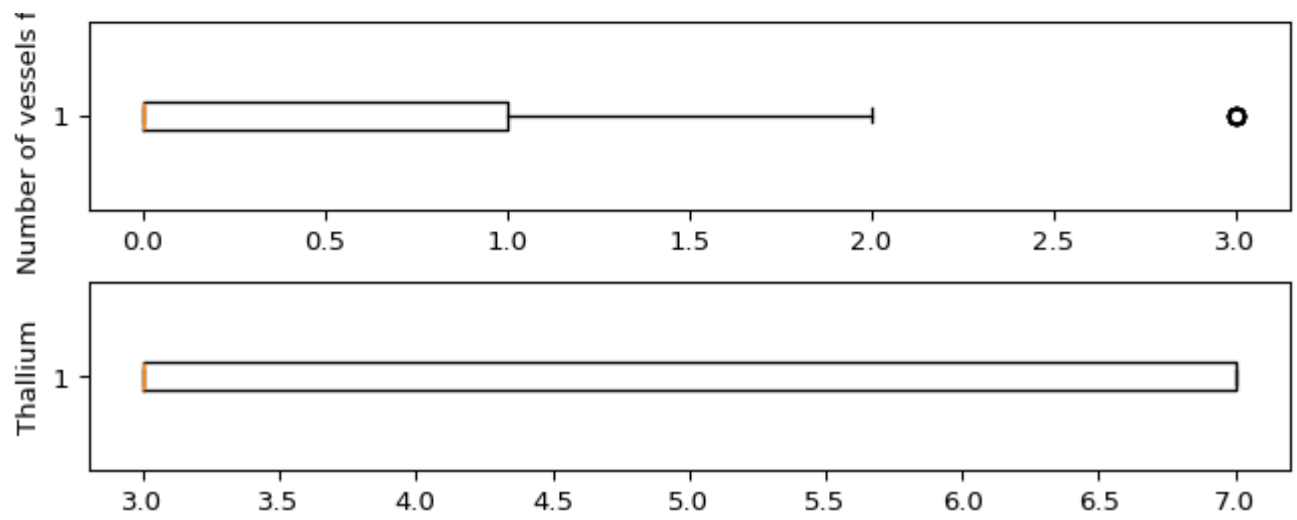
In [22]:
```
df.describe()
```

Out[22]:

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	o
count	270.000000	270.000000	270.000000	270.000000	270.000000	270.000000	270.000000	270.000000	270.000000	270.00000	270.000000	27
mean	54.433333	0.677778	3.174074	131.344444	249.659259	0.148148	1.022222	149.677778	0.329630	1.05000	1.585185	
std	9.109067	0.468195	0.950090	17.861608	51.686237	0.355906	0.997891	23.165717	0.470952	1.14521	0.614390	
min	29.000000	0.000000	1.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000	0.00000	1.000000	
25%	48.000000	0.000000	3.000000	120.000000	213.000000	0.000000	0.000000	133.000000	0.000000	0.00000	1.000000	
50%	55.000000	1.000000	3.000000	130.000000	245.000000	0.000000	2.000000	153.500000	0.000000	0.80000	2.000000	
75%	61.000000	1.000000	4.000000	140.000000	280.000000	0.000000	2.000000	166.000000	1.000000	1.60000	2.000000	
max	77.000000	1.000000	4.000000	200.000000	564.000000	1.000000	2.000000	202.000000	1.000000	6.20000	3.000000	

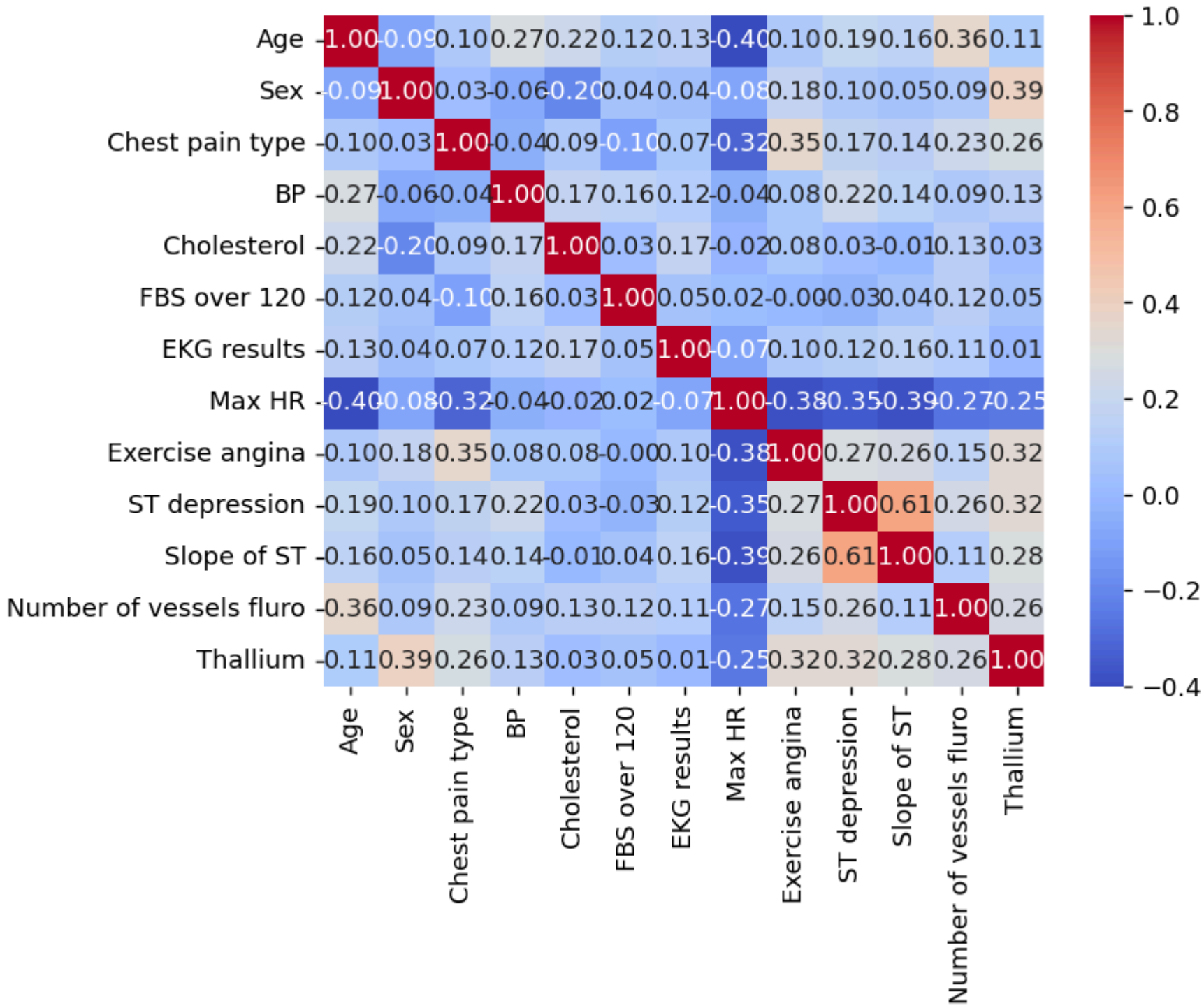
In [23]:
```
numeric_df = df.select_dtypes(include='number')

fig, axs = plt.subplots(len(numeric_df.columns), 1, figsize=(7, 18), dpi=95)

for i, col in enumerate(numeric_df.columns):
    axs[i].boxplot(numeric_df[col].dropna(), vert=False)
    axs[i].set_ylabel(col)

plt.tight_layout()
plt.show()
```


Age
Sex
Chest pain type
BP
Cholesterol
FBS over 120
EKG results
Max HR
Exercise angina
ST depression
Slope of ST



In [24]:
```
q1, q3 = np.percentile(df["Cholesterol"].dropna(), [25, 75])
iqr = q3 - q1

lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr

clean_df = df[(df["Cholesterol"] >= lower) & (df["Cholesterol"] <= upper)
]
```

In [25]:
```
corr = df.select_dtypes(include='number').corr()

plt.figure(dpi=130)
sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm')
plt.show()
```



In [26]:
```
X = df.select_dtypes(include='number')
X_normalized = MinMaxScaler().fit_transform(X)
print(X_normalized[:5])
```

```
[[0.85416667 1.         1.         0.33962264 0.44748858 0.
  1.         0.29007634 0.         0.38709677 0.5        1.
  0.        ]
 [0.79166667 0.         0.66666667 0.19811321 1.         0.
  1.         0.67938931 0.         0.25806452 0.5        0.
  1.        ]
 [0.58333333 1.         0.33333333 0.28301887 0.30821918 0.
  0.         0.53435115 0.         0.0483871  0.         0.
  1.        ]
 [0.72916667 1.         1.         0.32075472 0.31278539 0.
  0.         0.25954198 1.         0.03225806 0.5        0.33333333
  1.        ]
 [0.9375     0.         0.33333333 0.24528302 0.32648402 0.
  1.         0.38167939 1.         0.03225806 0.         0.33333333
  0.        ]]
```

In [27]:
```
X_standardized = StandardScaler().fit_transform(X)
print(X_standardized[:5])
```

```
[[ 1.71209356  0.6894997   0.87092765 -0.07540984  1.40221232 -0.41702883
   0.98166365 -1.75920811 -0.7012223   1.18101235  0.67641928  2.47268219
  -0.87570581]
 [ 1.38213977 -1.45032695 -0.18355874 -0.91675934  6.0930045  -0.41702883
   0.98166365  0.44640927 -0.7012223   0.48115318  0.67641928 -0.71153494
   1.18927733]
 [ 0.2822938   0.6894997  -1.23804513 -0.41194964  0.21982255 -0.41702883
  -1.02628472 -0.37529132 -0.7012223  -0.65611797 -0.95423434 -0.71153494
   1.18927733]
 [ 1.05218598  0.6894997   0.87092765 -0.18758978  0.25858943 -0.41702883
  -1.02628472 -1.93219771  1.4260813  -0.74360037  0.67641928  0.34987077
   1.18927733]
 [ 2.15203195 -1.45032695 -1.23804513 -0.63630951  0.37489006 -0.41702883
   0.98166365 -1.24023932  1.4260813  -0.74360037 -0.95423434  0.34987077
  -0.87570581]]
```